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Chapter

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Reflected Backward SDEs and American Options

N. El Karoui, E. Pardoux and M.C. Quenez

1 Introduction

We have seen in El Karoui & Quenez (1997) that the pricing of European contingent claims, even in imperfect markets, can be formulated in terms of backward stochastic differential equations. However, the case of American options has not been considered. In this article, we will see that the price of an American option corresponds to the solution of a new type of backward equation called reflected BSDEs. The solution of such an equation is forced to stay above a given stochastic process, called the obstacle. An increasing process is introduced which pushes the solution upwards, so that it may remain above the obstacle. Recall that by definition the price of an American option is constrained to be greater than the payoff of the option (which corresponds to the obstacle). Furthermore, in a perfect market, it is well known (see Bensoussan (1984), Karatzas & Shreve (1995) and Karatzas (1988)) that such options cannot be perfectly hedged by a portfolio; in this case, the price process corresponds to the minimal ‘superhedging’ strategy for the option, that is a strategy with a so-called ‘tracking’ error which is an increasing process. We will see that this property can be generalized to imperfect markets.

In this article, we will refer often to the previous one by El Karoui & Quenez (1997) concerning the different results on BSDEs and also the notation. The problem is formulated in detail in Section 2. We show that solutions of reflected BSDEs satisfy quite similar properties as in the classical case. First, we prove a comparison theorem for non-reflected BSDEs, from which we deduce the uniqueness of the solution of a reflected BSDE. We give also some a priori estimates on the spread of the solutions of two RBSDEs, which are very similar to the classical case. Then, we show the existence of a solution. The first step is to show the existence of the solution of a reflected equation whose generator is a given process. For classical BSDEs, this case is simply solved by using the representation theorem. In the case of a reflected BSDE, the solution is given by the value function of an optimal stopping time control problem (see Subsection 3.2). The second step is to use the a priori estimates so that a fixed point theorem can be applied. A second method to show the existence is based on the approximation by penalized classical BSDEs. This point of view is very classical in analysis for solving some PDEs.

The different properties concerning reflected BSDEs and optimization problems are studied in Section 3. First, we show that the solution can be associated with a classical deterministic Skohorod problem. From this, it is easy to show that the increasing process of the RBSDE can be expressed as an infimum. Furthermore, we state that the solution of the RBSDE corresponds to the value of an optimal stopping time problem (which depends on the solution itself). Finally, we study the case of reflected BSDE with concave or convex coefficients. Recall that in the case of a classical concave BSDE, the solution corresponds to the value function of an optimization problem (see El Karoui & Quenez (1997)). Similarly, the solution of a concave reflected BSDE is shown to be the value function of a mixed optimal stopping/optimal stochastic control problem. Furthermore, we have a separation theorem concerning the optimal stopping time and the optimal control. In Section 4, we study reflected BSDEs in a Markovian framework. Recall that it has been noticed in Pardoux & Peng (1992) that solutions of BSDEs are naturally connected with viscosity solutions of possibly degenerate parabolic PDEs. In this article, we show that, provided the problem is formulated within a Markovian framework, the solution of the reflected BSDE provides a probabilistic representation for the unique viscosity solution of an obstacle problem for a nonlinear parabolic partial differential equation.

Finally, we give some applications of reflected BSDEs to finance. We show that the classical results concerning the pricing of American options in a perfect market still hold in constrained cases. Recall that under some constraints such as a higher interest rate for borrowing, the strategies wealth-portfolio satisfy nonlinear convex stochastic differential equations. The price of a European option corresponds to the solution of a classical convex BSDE (see El Karoui & Quenez 1997). In the case of an American option, using the results of Section 3 on concave or convex reflected BSDEs, we show that the price process is a solution of a convex reflected BSDE; furthermore, it is equal to the so-called ‘upper price’ defined as the smallest price which allows us to construct a ‘superhedging’ strategy for the option.

2 Reflected BSDE, Comparison Theorem, Existence and Uniqueness

Put $\mathcal{S}^2 = \{\varphi \text{ predictable; } \mathbf{E}(\sup_{0 \leq t \leq T} |\varphi_t|^2) < +\infty\}$. The processes belonging to $\mathcal{S}^2$ will be called bounded in $\mathbb{L}^2$.

The problem can be formulated as follows:

We are given three objects:

- a terminal value $\xi \in \mathbb{L}^2$;
- a standard generator f ;

- an ‘obstacle’ $\{S_t, 0 \leq t \leq T\}$, which is a continuous adapted real valued process bounded in $\mathbb{L}^2$.
We shall always assume that $S_T \leq \xi$ a.s.
Each triple (ξ, f, S) satisfying the above assumptions is called a set of standard data.

Let us now introduce our reflected BSDE. The solution of our RBSDE is a triple $\{Y_t, Z_t, K_t\}$, $0 \leq t \leq T$ of $\mathcal{F}_t$ -progressively measurable processes taking values in $\mathbb{R}$, $\mathbb{R}^n$ and $\mathbb{R}_+$ respectively, and satisfying:

- (i) $Z \in \mathbb{H}^2$, $Y \in \mathcal{S}^2$ and $K_T \in \mathbb{L}^2$.
- (ii) $Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds + K_T - K_t - \int_t^T (Z_s dB_s)$, $0 \leq t \leq T$;
- (iii) $Y_t \geq S_t$, $0 \leq t \leq T$;
- (iv) $\{K_t\}$ is continuous and increasing, $K_0 = 0$, and $\int_0^T (Y_t - S_t) dK_t = 0$.

Note that from (ii) and (iv) it follows that $\{Y_t\}$ is continuous.

Intuitively, dK_t/dt represents the amount of ‘push upwards’ that we add to $-dY_t/dt$, so that the constraint (i) is satisfied. Condition (iv) says that the push is minimal, in the sense that we push only when the constraint is saturated, i.e. when $Y_t = S_t$. We will see that this minimality property can be derived from a comparison theorem.

Indeed, as in the classical case, we have a comparison theorem very similar to the previous one, from which the uniqueness of the solution of the RBSDE follows. The proof of the comparison theorem for RBSDEs differs from the proof for classical BSDEs because the difference of the solutions of two RBSDEs is no longer a solution of a RBSDE. However, we will see that the properties (iv) of the increasing process K allow us to derive the desired result.

Theorem 2.1 *Let (ξ, f, S) and (ξ', f', S') be two sets of standard data, and suppose in addition that*

- $\xi \leq \xi'$ a.s.;
- $f(t, y, z) \leq f'(t, y, z)$ $dP \times dt$ a.e., $\forall (y, z) \in \mathbb{R} \times \mathbb{R}^d$;
- $S_t \leq S'_t$, $0 \leq t \leq T$, a.s.

Let (Y, Z, K) be a solution of the RBSDE with data (ξ, f, S) , and (Y', Z', K') a solution of the RBSDE with data (ξ', f', S') . Then

$$Y_t \leq Y'_t, 0 \leq t \leq T, \text{ a.s.}$$

Proof Applying Itô's formula to $|(Y_t - Y'_t)^+|^2$, and taking the expectation, we have:

$$\begin{aligned} E|(Y_t - Y'_t)^+|^2 &+ E \int_t^T \mathbf{1}_{\{Y_s > Y'_s\}} |Z_s - Z'_s|^2 ds \\ &\leq 2E \int_t^T (Y_s - Y'_s)^+ [f(s, Y_s, Z_s) - f'(s, Y'_s, Z'_s)] ds \\ &\quad + 2E \int_t^T (Y_s - Y'_s)^+ (dK_s - dK'_s). \end{aligned}$$

Since¹ on $\{Y_t > Y'_t\}$, $Y_t > S'_t \geq S_t$,

$$\begin{aligned} \int_t^T (Y_s - Y'_s)^+ (dK_s - dK'_s) &= - \int_t^T (Y_s - Y'_s)^+ dK'_s \\ &\leq 0 \end{aligned}$$

Now, since f is Lipschitz with constant C , it follows that

$$\begin{aligned} E|(Y_t - Y'_t)^+|^2 &+ E \int_t^T \mathbf{1}_{\{Y_s > Y'_s\}} |Z_s - Z'_s|^2 ds \\ &\leq 2E \int_t^T (Y_s - Y'_s)^+ [f(s, Y_s, Z_s) - f(s, Y'_s, Z'_s)] ds \\ &\leq 2CE \int_t^T (Y_s - Y'_s)^+ (|Y_s - Y'_s| + |Z_s - Z'_s|) ds \\ &\leq E \int_t^T \mathbf{1}_{\{Y_s > Y'_s\}} |Z_s - Z'_s|^2 ds + \bar{C}E \int_t^T |(Y_s - Y'_s)^+|^2 ds \end{aligned}$$

(we have used the fact that $2Cyz \leq C^2y^2 + z^2$). Hence

$$E|(Y_t - Y'_t)^+|^2 \leq \bar{C}E \int_t^T |(Y_s - Y'_s)^+|^2 ds,$$

and from Gronwall's Lemma, $(Y_t - Y'_t)^+ = 0$, $0 \leq t \leq T$. $\square$

From the proof of the comparison theorem, we derive the minimality property of the increasing process K . More precisely,

Corollary 2.2 *Let (Y, Z, K) be a solution of the RBSDE with standard data (ξ, f, S) and let (Y', Z', K') be a triple of processes satisfying (i), (ii), (iii) (associated with the same data (ξ, f, S)) but with K' a general continuous increasing process with $K'_0 = 0$ but not necessarily $\int_0^T (Y'_t - S_t) dK'_t = 0$. Then*

$$Y_t \leq Y'_t, \quad 0 \leq t \leq T, \quad \text{a.s.}$$

Also, we deduce immediately the following uniqueness result from the comparison theorem with $\xi' = \xi$, $f' = f$ and $S' = S$:

¹Notice that the property $\int_0^T (Y'_t - S'_t) dK'_t = 0$ is not used in the proof. Hence, the result still holds if (Y', Z', K') satisfies only (i), (ii), (iii) with K' a general continuous increasing process with $K'_0 = 0$.

Corollary 2.3 *Let (ξ, f, S) be a set of standard data. There exists at most one progressively measurable triple $\{(Y_t, Z_t, K_t), 0 \leq t \leq T\}$ of the RBSDE associated with data (ξ, f, S) .*

It remains to show the existence of a solution of the RBSDE. As in the case of a classical BSDE, we derive some a priori estimates concerning the spread between the solutions of two different RBSDEs very similar to those obtained in the classical case (see the proposition on ‘a priori estimates’ in El Karoui & Quenez (1997)); then, once we have proved the existence for a generator $f(t)$ which does not depend on y, z , we derive (using the a priori estimates) the existence of the solution of a RBSDE by a fixed point theorem. More precisely,

Theorem 2.4

- (*A priori estimates*) Let $((\xi^i, f^i, S^i); i = 1, 2)$ be two sets of standard data of an RBSDE with the same obstacle $S^1 = S^2$, and $((Y^i, Z^i, K^i); i = 1, 2)$ the solutions of the associated RBSDE.

Then, the a priori estimates still hold.

- (*Existence*) Let (ξ, f, S) be a set of standard data. Then, the RBSDE defined by (i), (ii), (iii) and (iv) has a unique solution $\{(Y_t, Z_t, K_t); 0 \leq t \leq T\}$.

Sketch of the Proof The a priori estimates can be derived by using the same arguments as in the classical case; indeed, as in the proof of the comparison theorem, the supplementary terms concerning the increasing processes K^1 and K^2 which appear in the inequalities do not create any difficulty because they are negative (by property (iv)).

Uniqueness has already been shown. Notice that instead of saying that a triple $\{(Y_t, Z_t, K_t); 0 \leq t \leq T\}$ of $\mathbb{R} \times \mathbb{R}^d \times \mathbb{R}_+$ -valued progressively measurable processes is a solution of our RBSDE, we could say that a pair $\{(Y_t, Z_t); 0 \leq t \leq T\}$ of $\mathbb{R} \times \mathbb{R}^d$ -valued progressively measurable processes satisfying (i) and (iii) is a solution of our RBSDE, meaning that, if $\{K_t, 0 \leq t \leq T\}$ is defined by (ii), then the pair (Y, K) satisfies moreover (iv). In that sense, it follows from Corollary 2.3 that there exists at most one pair $\{(Y_t, Z_t); 0 \leq t \leq T\}$ of progressively measurable processes which solves the RBSDE.

It remains now to prove existence. In the case where f does not depend on (y, z) , the existence of the solution of the RBSDE will be shown in the next section (see Proposition 3.4) by using a result of the optimal stopping problem.

Suppose now that the generator $f(t, y, z)$ is general. As in the proof of the existence theorem for classical BSDEs, we use a fixed point theorem, for

the mapping ϕ defined on $\mathbf{H}_\beta^2 \times \mathbf{H}_\beta^2$ which maps (y_t, z_t) into the pair (Y_t, Z_t) , where (Y, Z, K) is the solution of the RBSDE associated with the generator $f(t, y_t, z_t)$ (which does not depend on Y, Z). In other words, (Y_t, Z_t, K_t) is the solution of the RBSDE associated with data $(\xi, f(t, y_t, z_t), S)$. It remains to show that for a good choice of β , ϕ is a contraction. Let $(y^1, z^1), (y^2, z^2)$ be two elements of $\mathbf{H}_\beta^2 \times \mathbf{H}_\beta^2$ and let $(Y^1, Z^1, K^1), (Y^2, Z^2, K^2)$ be the associated solutions. Then, by the a priori estimates, we find that for β sufficiently large, there exists a positive constant $\epsilon < 1$ such that the following inequality holds

$$\|Y^1 - Y^2\|_\beta^2 + \|Z^1 - Z^2\|_\beta^2 < \epsilon(\|y^1 - y^2\|_\beta^2 + \|z^1 - z^2\|_\beta^2),$$

for any elements $(y^1, z^1), (y^2, z^2)$ of $\mathbf{H}_\beta^2 \times \mathbf{H}_\beta^2$ and their associated solutions $(Y^1, Z^1, K^1), (Y^2, Z^2, K^2)$. Hence, ϕ is a contraction and there exists a fixed point (Y, Z) which is the unique solution of the RBSDE. The associated increasing process K is then defined by condition (ii). $\square$

To show existence and uniqueness, another point of view is to approximate the solution of the RBSDE by classical BSDEs which are obtained by penalizing the solution if it is smaller than the obstacle. These techniques are often used in analysis to solve some PDEs. More precisely,

Proposition 2.5 *For each $n \in \mathbb{N}$, let $\{(Y_t^n, Z_t^n); 0 \leq t \leq T\}$ denote the unique pair of square integrable $\mathcal{F}_t$ -progressively measurable processes satisfying*

$$Y_t^n = \xi + \int_t^T f(s, Y_s^n, Z_s^n) ds + n \int_t^T (Y_s^n - S_s)^- ds - \int_t^T (Z_s^n, dB_s). \quad (2.1)$$

Then, the increasing sequence Y_t^n converges a.s. and in S^2 to Y_t , the sequence Z^n converges to Z in $\mathbf{H}^2$ and K^n converges to K in S^2 as n tends to infinity, where K^n is defined by

$$K_t^n = n \int_0^t (Y_s^n - S_s)^- ds, \quad 0 \leq t \leq T.$$

The proof is based on the estimations K^n , and the difficulties come from the fact that the almost sure convergence of the increasing sequence Y_t^n is not sufficient and it is necessary to prove the uniform convergence of the different processes. For details, one is referred to El Karoui, Kapoudjian, Pardoux, Peng & Quenez (1995).

3 Reflected BSDEs, the Skohorod Problem and Optimal Stopping Time/Control Problems

In this section, we study the different control problems which are associated with a RBSDE. We first show that the solution can be associated with a

classical deterministic Skohorod problem. From this, it is easy to deduce that the increasing process can be written as a supremum.

3.1 RBSDEs and the Skohorod problem

Notice that in a deterministic framework, the formulation of the RBSDE corresponds to the Skohorod problem. Consequently, we will be able to apply some well known properties of the Skohorod problem. Recall the Skohorod lemma (see for example El Karoui & Chayelot-Maurel (1978) and Revuz & Yor (1994) p. 229).

Proposition 3.1 (Skohorod lemma)

Let x be a real-valued continuous function on $[0, \infty[$ such that $x_0 \geq 0$. There exists a unique pair (y, k) of functions on $[0, \infty[$ such that

$$(a) \quad y = x + k$$

$$(b) \quad y \text{ is positive,}$$

$$(c) \quad \{k_t\} \text{ is continuous and increasing, } k_0 = 0, \text{ and } \int_0^\infty y_t dk_t = 0.$$

The pair (y, k) is said to be the solution of the Skohorod problem. The function k is moreover given by

$$k_t = \sup_{s \leq t} x_s^-.$$

Now, our problem involves a Skohorod problem and consequently, the increasing process can be written as a supremum. More precisely,

Proposition 3.2 Let $\{(Y_t, Z_t, K_t), 0 \leq t \leq T\}$ be a solution of the above RBSDE satisfying conditions (ii) to (iv). Then for each $t \in [0, T]$,

$$K_T - K_t = \sup_{t \leq u \leq T} \left(\xi + \int_u^T f(s, Y_s, Z_s) ds - \int_u^T (Z_s dB_s) - S_u \right)^-. \quad (3.1)$$

Proof Notice that $(Y_{T-t}(\omega) - S_{T-t}(\omega), K_{T-t}(\omega) - K_T(\omega), 0 \leq t \leq T)$ is the solution of a Skohorod problem. Applying the Skohorod lemma with

$$\begin{aligned} x_t &= \left(\xi + \int_{T-t}^T f(s, Y_s, Z_s) ds - \int_{T-t}^T (Z_s, dB_s) - S_{T-t} \right)(\omega), \\ k_t &= (K_T - K_{T-t})(\omega) \\ y_t &= (Y_{T-t} - S_{T-t})(\omega), \end{aligned}$$

we derive the desired result. $\square$

It is not at all clear from (3.1) that $\{K_t\}$ will be $\mathcal{F}_t$ -adapted. The adaptedness of (Y, K) will come from the adjustment of the process Z . In other words, Z is the process which has the effect of making (Y, K) adapted.

In the next section we show that the solution of the RBSDE can be associated with a stopping time optimal problem.

3.2 RBSDEs and stopping time optimal problem

In the following proposition, we show that the square-integrable solution Y_t of the RBSDE corresponds to the value of a stopping time optimal problem.

Proposition 3.3 *Let $\{(Y_t, Z_t, K_t), 0 \leq t \leq T\}$ be a solution of the above RBSDE satisfying conditions (i) to (iv); then for each $t \in [0, T]$,*

$$Y_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} E \left[\int_t^v f(s, Y_s, Z_s) ds + S_v \mathbf{1}_{\{v < T\}} + \xi \mathbf{1}_{\{v = T\}} | \mathcal{F}_t \right] \quad (3.2)$$

where $\mathcal{T}$ is the set of all stopping times dominated by T , and

$$\mathcal{T}_t = \{v \in \mathcal{T}; t \leq v \leq T\}.$$

Proof Let $v \in \mathcal{T}_t$. From (i) and (iv), we may take the conditional expectation in (ii) written between times t and v , hence

$$\begin{aligned} Y_t &= E \left[\int_t^v f(s, Y_s, Z_s) ds + Y_v + K_v - K_t | \mathcal{F}_t \right] \\ &\geq E \left[\int_t^v f(s, Y_s, Z_s) ds + S_v \mathbf{1}_{\{v < T\}} + \xi \mathbf{1}_{\{v = T\}} | \mathcal{F}_t \right]. \end{aligned}$$

We now choose an optimal element of $\mathcal{T}_t$ in order to get the reversed inequality. Let

$$D_t = \inf\{t \leq u \leq T; Y_u = S_u\},$$

with the convention that $D_t = T$ if $Y_u > S_u$, $t \leq u \leq T$. Now the condition $\int_0^T (Y_t - S_t) dK_t = 0$, and the continuity of K implies that

$$K_{D_t} - K_t = 0.$$

It follows that

$$Y_t = E \left[\int_t^{D_t} f(s, Y_s, Z_s) ds + S_{D_t} \mathbf{1}_{\{D_t < T\}} + \xi \mathbf{1}_{\{D_t = T\}} | \mathcal{F}_t \right].$$

Hence the result follows. $\square$

It is clear from the Proposition 3.3 above that in the case where f is a given stochastic process, the solution $\{Y_t; 0 \leq t \leq T\}$ of the RBSDE is the value function of an optimal stopping time problem (which does not depend on (Y, Z)). From this fact, this value function seems to be a natural candidate to be the solution of the RBSDE in this case. We will show in the next section that it is indeed the solution.

3.3 Existence in the case of a given generator

By the results of optimal stopping theory, we derive the following existence result:

Proposition 3.4 *Suppose that $f(t)$ is a given predictable stochastic process in $\mathbf{H}^2$. Then, the RBSDE associated with (ξ, f, S) has a unique solution.*

Proof From Proposition 3.3, it is natural to introduce the process $\{Y_t; 0 \leq t \leq T\}$ defined by

$$Y_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} E \left[\int_t^v f(s) ds + S_v \mathbf{1}_{\{v < T\}} + \xi \mathbf{1}_{\{v = T\}} | \mathcal{F}_t \right], \quad 0 \leq t \leq T.$$

It is clear that $Y_t \geq S_t$. Also, optimal stopping theory (cf. El Karoui 1981, Karatzas & Shreve 1995) shows that there exists a continuous version of Y_t which can be written as the solution of the RBSDE (see El Karoui, Kapoudjian, Pardoux, Peng & Quenez 1995). $\square$

In the next section, we study the case of an RBSDE associated with a linear generator and show that, as in the case of a generator defined as a given process, the solution $\{Y_t; 0 \leq t \leq T\}$ of the RBSDE is the value function of an optimal stopping time problem (which does not depend on the solution (Y, Z)).

3.4 Linear RBSDE

In the linear case, that is when $f(t, y, z)$ is a linear function of (y, z) , Proposition 3.3 shows that, as for a classical linear BSDE, the solution can be written using the adjoint process.

Proposition 3.5 *Let (β_t, γ_t) be a bounded $(\mathbf{R}, \mathbf{R}^n)$ -valued predictable vector process, and φ_t be an element of $\mathbf{H}_T^2(\mathbf{R})$. Let f be the standard generator defined by*

$$f(t, y, z) = \varphi_t + \beta_t y + \gamma_t^* z.$$

Let $(\Gamma_{t,s}; t \leq s \leq T)$ be the adjoint process satisfying the linear SDE:

$$d\Gamma_{t,s} = \Gamma_{t,s}(\beta_s ds + \gamma_s^* dB_s); \quad \Gamma_{t,t} = 1.$$

Then the unique solution $\{Y_t, Z_t, K_t; 0 \leq t \leq T\}$ of the RBSDE with generator f satisfies, for each $0 \leq t \leq T$,

$$Y_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} E \left[\Gamma_{t,v} \xi \mathbf{1}_{\{v = T\}} + \Gamma_{t,v} S_v \mathbf{1}_{\{v < T\}} + \int_t^v \Gamma_{t,s} \varphi_s ds | \mathcal{F}_t \right],$$

which can also be written

$$Y_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} X_t(v, \tilde{S}_v)$$

where $(X_t(v, \tilde{S}_v), 0 \leq t \leq v)$ is the solution associated with the BSDE associated with generator f , terminal time v and terminal condition $\tilde{S}_v$ where

$$\tilde{S}_t = \xi \mathbf{1}_{\{t=T\}} + S_t \mathbf{1}_{\{t < T\}}.$$

Furthermore, the stopping time $D_t = \inf\{t \leq s \leq T; Y_s = S_s\}$ is optimal.

Proof Let us denote the process $\Gamma_{0,s}$ by Γ_s . It follows from Itô's formula that

$$Y_t \Gamma_t = \xi \Gamma_T + \int_t^T \Gamma_s \varphi_s ds + \int_t^T \Gamma_s dK_s - \int_t^T \Gamma_s (Z_s + Y_s \gamma_s)^* dB_s.$$

Let $(Y'_t, Z'_t, K'_t) = (Y_t \Gamma_t, \Gamma_t(Z_t + Y_t \gamma_t), \int_0^t \Gamma_s dK_s)$, $0 \leq t \leq T$. This triplet solves the RBSDE with final condition $\xi \Gamma_T$, coefficient $\{\Gamma_t \varphi_t, 0 \leq t \leq T\}$ and obstacle $\Gamma_t S_t$, without Condition (i) in Section 2.

Also we only have that

$$E[(\xi \Gamma_T)^{2-\varepsilon} + \int_0^T (\varphi_t \Gamma_t)^{2-\varepsilon} dt] < \infty$$

for each $\varepsilon > 0$, and not for $\varepsilon = 0$; the argument leading to Proposition 3.3 is still valid here, hence

$$\Gamma_t Y_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} E \left[\Gamma_T \xi \mathbf{1}_{\{v=T\}} + \Gamma_v S_v \mathbf{1}_{\{v < T\}} + \int_t^v \Gamma_{t,s} \varphi_s ds / \mathcal{F}_t \right],$$

and D_t is optimal. Consider now the solution $(X_t(v, \tilde{S}_v), 0 \leq t \leq v)$ as the solution associated with the BSDE associated with generator f , terminal time v and terminal condition $\tilde{S}_v = \xi \mathbf{1}_{\{v=T\}} + S_v \mathbf{1}_{\{v < T\}}$. Then, by classical results on linear BSDEs, it follows that

$$X_t(v, \tilde{S}_v) = E \left[\Gamma_{t,v} \tilde{S}_v + \int_t^v \Gamma_{t,s} \varphi_s ds / \mathcal{F}_t \right],$$

and hence, Y_t can also be written

$$Y_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} X_t(v, \tilde{S}_v).$$

Furthermore, the fact that D_t is optimal follows from the argument given in the proof of Proposition 3.3. $\square$

In the next section, we will see that in the case where f is a concave (or convex) function of (y, z) , we have that $\{Y_t, 0 \leq t \leq T\}$ is the value function of a mixture of an optimal stopping time problem and a 'classical' optimal stochastic control problem.

3.5 Concave RBSDEs and the optimal stopping time/control problem

We now suppose that $f(t, y, z)$ is a concave function of (y, z) . We use the same notation as El Karoui & Quenez (1997, Section 3) on concave BSDEs. Recall that the conjugate function of f is denoted by $F(t, \beta, \gamma)$ and that $\mathcal{A}$ denotes the set of bounded progressively measurable $\mathbb{R} \times \mathbb{R}^d$ -valued processes $\{(\beta_t, \gamma_t); 0 \leq t \leq T\}$ which are such that

$$E \int_0^T F(t, \beta_t, \gamma_t)^2 dt < \infty.$$

To each $(\beta, \gamma) \in \mathcal{A}$, we associate the unique solution $\{(Y_t^{\beta, \gamma}, Z_t^{\beta, \gamma}, K_t^{\beta, \gamma}); 0 \leq t \leq T\}$ of the linear RBSDE with generator $f^{\beta, \gamma}(t, y, z) = F(t, \beta_t, \gamma_t) + \beta_t y + \gamma_t^* z$. We shall denote by $\{(Y_t, Z_t, K_t); 0 \leq t \leq T\}$ the unique solution of the RBSDE with generator $f(t, y, z)$.

Recall that by the conjugacy relation,

$$f(t, Y_t, Z_t) = \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} f^{\beta, \gamma}(t, Y_t, Z_t) \quad dt \times dP \text{ a.e.}$$

and hence, by the comparison theorem (for RBSDEs) 2.1,

$$Y_t \leq Y_t^{\beta, \gamma}, \quad \forall (\beta, \gamma) \in \mathcal{A}.$$

Furthermore, as in the case of a classical concave BSDE (see El Karoui & Quenez 1997), by a selection measurable theorem, there exists $(\bar{\beta}, \bar{\gamma}) \in \mathcal{A}$ such that

$$f(t, Y_t, Z_t) = f^{\bar{\beta}, \bar{\gamma}}(t, Y_t, Z_t) \quad dt \times dP \text{ a.e.}, \quad (3.3)$$

hence,

$$(Y_t, Z_t, K_t) = (Y_t^{\bar{\beta}, \bar{\gamma}}, Z_t^{\bar{\beta}, \bar{\gamma}}, K_t^{\bar{\beta}, \bar{\gamma}}), \quad 0 \leq t \leq T, \quad \text{a.s.}$$

It follows that

$$Y_t = Y_t^{\bar{\beta}, \bar{\gamma}} \geq \inf_{(\beta, \gamma) \in \mathcal{A}} Y_t^{\beta, \gamma},$$

Consequently, as in the case of a concave classical BSDE, we have

Proposition 3.6 *For each t , Y_t can be written as the value function of an optimal problem given by*

$$Y_t = \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} Y_t^{\beta, \gamma}.$$

Furthermore, the control $(\bar{\beta}, \bar{\gamma})$ is optimal, that is

$$Y_t = Y_t^{\bar{\beta}, \bar{\gamma}}.$$

Now, by the previous results on linear RBSDEs (see Proposition 3.5), we know that for each $(\beta, \gamma) \in \mathcal{A}$, the process $Y_t^{\beta, \gamma}$ can be written

$$Y_t^{\beta, \gamma} = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} X_t^{\beta, \gamma}(v, \tilde{S}_v) \quad (3.4)$$

where $X_t^{\beta, \gamma}(v, \tilde{S}_v)$ is the solution of the BSDE associated with generator $f^{\beta, \gamma}$, terminal time v and terminal condition $\tilde{S}_v$. Recall that

$$X_t^{\beta, \gamma}(v, \tilde{S}_v) = E[\Gamma_{t,v}^{\beta, \gamma} \tilde{S}_v + \int_t^v \Gamma_{t,s}^{\beta, \gamma} F(s, \beta_s, \gamma_s) ds / \mathcal{F}_t].$$

By Proposition 3.6 and equation (3.4), we deduce that Y_t can be written as the value function of an optimal-stopping time/control problem, that is

Theorem 3.7 *For each t , Y_t is the value function of a minimax problem, that is*

$$Y_t = \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} \operatorname{ess\,sup}_{v \in \mathcal{T}_t} X_t^{\beta, \gamma}(v, \tilde{S}_v).$$

Moreover, $\operatorname{ess\,inf}$ and $\operatorname{ess\,sup}$ can be interchanged, that is Y_t is also the value function of a maxmin problem given by

$$Y_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} X_t^{\beta, \gamma}(v, \tilde{S}_v).$$

and the triple $(\bar{\beta}, \bar{\gamma}, D_t)$, where $D_t = \inf\{t \leq s \leq T; Y_s = S_s\}$, is optimal.

Remark Notice that the optimal control $(\bar{\beta}, \bar{\gamma})$ does not depend on D_t and the definition of D_t does not depend on $(\bar{\beta}, \bar{\gamma})$. The optimal control $(\bar{\beta}, \bar{\gamma}, D_t)$ satisfies a so-called ‘separation’ principle.

Proof The first equality follows directly from Proposition 3.6 and equation (3.4). We have to prove $\operatorname{ess\,inf}$ and $\operatorname{ess\,sup}$ can be interchanged. We certainly have

$$\begin{aligned} Y_t &= \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} \operatorname{ess\,sup}_{v \in \mathcal{T}_t} X_t^{\beta, \gamma}(v, \tilde{S}_v) \\ &\geq \operatorname{ess\,sup}_{v \in \mathcal{T}_t} \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} X_t^{\beta, \gamma}(v, \tilde{S}_v). \end{aligned}$$

On the other hand, by Proposition 3.5, the stopping time $D_t^{\beta, \gamma}$, given by $D_t^{\beta, \gamma} = \inf\{t \leq s \leq T; Y_s^{\beta, \gamma} = S_s\}$, is optimal for the optimization problem

$$\sup_{v \in \mathcal{T}_t} X_t^{\beta, \gamma}(v, \tilde{S}_v),$$

and consequently,

$$\begin{aligned} Y_t &= \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} X_t^{\beta, \gamma}(D_t^{\beta, \gamma}, \tilde{S}_{D_t^{\beta, \gamma}}) \\ &\leq \operatorname{ess\,sup}_{v \in \mathcal{T}_t} \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} X_t^{\beta, \gamma}(v, \tilde{S}_v). \end{aligned}$$

The desired result follows.

Furthermore, by noticing that $D_t = D_t^{\bar{\beta}, \bar{\gamma}}$, it follows that

$$Y_t = Y_t^{\bar{\beta}, \bar{\gamma}} = X_t^{\bar{\beta}, \bar{\gamma}}(D_t, \tilde{S}_{D_t})$$

and hence, the triple $(\bar{\beta}, \bar{\gamma}, D_t)$ is optimal for the problem

$$Y_t = \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} \operatorname{ess\,sup}_{v \in \mathcal{T}_t} X_t^{\beta, \gamma}(v, \tilde{S}_v). \quad \square$$

Note that one has a similar representation of Y_t in the case where f is a convex function of (y, z) , with $\operatorname{ess\,inf}_{(\beta, \gamma)} \operatorname{ess\,sup}_v[\cdot]$ replaced by $\operatorname{ess\,sup}_{(\beta, \gamma)} \operatorname{ess\,sup}_v[\cdot]$.

4 Relation between a RBSDE and an Obstacle Problem for a Nonlinear Parabolic PDE

In this section, we will show that the reflected BSDE studied in the previous sections allows us to give a probabilistic representation of solutions of some obstacle problems for PDEs. For that purpose, we will put the RBSDE in a Markovian framework. The framework is the same as in the case of Markovian BSDEs and the assumptions on the state process and the coefficients b, σ, f and g are the same. Furthermore, the obstacle S_t is supposed to satisfy

$$S_s = h(s, S_s^{t,x})$$

where $h : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies

$$h(t, x) \leq K(1 + |x|^p), \quad t \in [0, T], \quad x \in \mathbb{R}^d. \quad (4.1)$$

We assume moreover that $h(T, x) \leq g(x)$, $x \in \mathbb{R}^d$.

We will denote by $(Y_s^{t,x}, Z_s^{t,x}, K_s^{t,x} \mid t \leq s \leq T)$ the solution of the RBSDE associated with obstacle S_s , generator $f(s, S_s^{t,x}, y, z)$ and terminal condition $g(S_T^{t,x})$. As in the case of a classical BSDE, we know that $Y_t^{t,x}$ is a deterministic function of (t, x) denoted by $u(t, x)$. We have the following theorem:

Theorem 4.1 *Suppose that the coefficients f, b, σ, g, h are jointly continuous in t and x . Then, the function $u(t, x)$ is the unique viscosity solution of the obstacle problem:*

$$\begin{cases} \min \left(u(t, x) - h(t, x), -\frac{\partial u}{\partial t}(t, x) - L_t u(t, x) - f(t, x, u(t, x), (\nabla u \sigma)(t, x)) \right) = 0, & (t, x) \in (0, T) \times \mathbb{R}^d \\ u(T, x) = g(x), \quad x \in \mathbb{R}^d; \end{cases} \quad (4.2)$$

where L_t is the infinitesimal operator associated with the Itô process $S^{t,x}$.

Sketch of the proof We are going to use the approximation of the RBSDE in Proposition 2.5 by penalization.

For each $(t, x) \in [0, T] \times \mathbb{R}^d$, $n \in \mathbb{N}^*$, let $\{({}^nY_s^{t,x}, {}^nZ_s^{t,x}), t \leq s \leq T\}$ denote the solution of the BSDE:

$$\begin{aligned} {}^nY_s^{t,x} &= g(S_T^{t,x}) + \int_s^T f(r, S_r^{t,x}, {}^nY_r^{t,x}, {}^nZ_r^{t,x}) dr \\ &\quad + n \int_s^T ({}^nY_r^{t,x} - h(r, S_r^{t,x}))^- - \int_s^T {}^nZ_r^{t,x} dB_r, \quad t \leq s \leq T. \end{aligned}$$

By the previous results on classical BSDEs in the Markovian case, we know that

$$u_n(t, x) = {}^nY_t^{t,x}, \quad 0 \leq t \leq T, \quad x \in \mathbb{R}^d,$$

is the viscosity solution of the parabolic PDE:

$$\begin{cases} \frac{\partial u_n}{\partial t}(t, x) + L_t u_n(t, x) + f_n(t, x, u_n(t, x), (\nabla u_n \sigma)(t, x)) = 0, & 0 \leq t \leq T; \\ u(T, x) = g(x); \end{cases}$$

where $f_n(t, x, r, p\sigma(t, x)) = f(t, x, r, p\sigma(t, x)) + n(r - h(t, x))^-$, and $x \in \mathbb{R}^d$.

But from the results on the approximation of the RBSDE (Proposition 2.5) by penalization, for each $0 \leq t \leq T$, $x \in \mathbb{R}^d$,

$$u_n(t, x) \uparrow u(t, x) \text{ as } n \rightarrow \infty.$$

Since u_n and u are continuous, it follows from Dini's theorem that the above convergence is uniform on compact sets. Using this property and classical techniques of the theory of viscosity solutions, it is possible to show that $u(t, x)$ is a viscosity solution of the obstacle problem. Concerning the proof of the uniqueness of the viscosity solution of the obstacle problem, one is referred to El Karoui, Kapoudjian, Pardoux, Peng & Quenez (1995). $\square$

In the next section, we are concerned with the application of RBSDEs to the pricing of American options.

5 Application of RBSDEs to Finance

We have seen in the study of classical concave and convex BSDEs and the 'application to European option pricing in the constrained case' (see El Karoui & Quenez 1997, Section 3) that in some constraint cases, the strategy wealth-portfolio (X_t, π_t) is a pair of adapted processes in $\mathbf{H}^2(\mathbb{R}) \times \mathbf{H}^2(\mathbb{R}^n)$ which satisfies the following BSDE

$$-dX_t = b(t, X_t, \pi_t)dt - \pi_t^* \sigma_t dW_t. \quad (5.1)$$

Here, b is an $\mathbb{R}$ -valued function defined on $[0, T] \times \Omega \times \mathbb{R} \times \mathbb{R}^n$ satisfying the standard hypotheses of a generator and convex with respect to x, π . Furthermore, we will suppose that the volatility matrix σ_t of the n risky assets

is invertible and that $(\sigma_t)^{-1}$ is uniformly bounded. The classical case is a special case corresponding to a linear functional

$$b(t, x, \pi) = -r_t x - \pi^* \sigma_t \theta_t,$$

where θ is the risk premium vector. We can suppose, without loss of generality, that $\sigma_t = \text{Id}$.

We are concerned with the problem of pricing an American contingent claim at each time t which consists of the selection of a stopping time $v \in \mathcal{T}_t$ and a payoff S_v on exercise if $v < T$ and ξ if $v = T$. Here, $(S_t, 0 \leq t \leq T)$ satisfies the assumptions of the 'obstacle'. Put

$$\tilde{S}_s = \xi \mathbf{1}_{\{s=T\}} + S_s \mathbf{1}_{\{v < T\}} \quad 0 \leq s \leq T.$$

The pricing method for pricing the American contingent claim is based on the same arguments as in the non constrained case (see Karatzas (1989) or Karatzas & Shreve (1995)). Fix $t \in [0, T]$. Suppose for the moment that the choice of $v \in \mathcal{T}_t$ has been made; then, there exists a unique strategy $(X_s(v, \tilde{S}_v), \pi_s(v, \tilde{S}_v)) \in \mathbb{H}^2 \times \mathbb{H}^2$, denoted also by (X_s^v, π_s^v) , which replicates $\tilde{S}_v$, i.e. is the solution of the classical BSDE associated with terminal time v , terminal condition $\tilde{S}_v$ and generator b :

$$-dX_s^v = b(s, X_s^v, \pi_s^v)ds - (\pi_s^v)^* dW_s, \quad 0 \leq s \leq v \quad X_v^v = \tilde{S}_v. \quad (5.2)$$

Then, the price of the American contingent claim $(\tilde{S}_s, 0 \leq s \leq T)$ at time t is given by the right-continuous adapted process X satisfying at each time t

$$X_t = \text{ess sup}_{v \in \mathcal{T}_t} X_t(v, \tilde{S}_v).$$

Recall that b is convex and let $B(t, \beta, \gamma)$ be the polar process associated with b . By the results of Section 3 on concave or convex classical BSDEs (see El Karoui & Quenez 1997), we know that X_t^v can be written as the maximum of ex-post prices over all feasible deflators

$$X_t(v, \tilde{S}_v) = \text{ess sup} \{ X_t^{\beta, \gamma}(v, \tilde{S}_v) \mid (\beta, \gamma) \in \mathcal{A} \},$$

where $\mathcal{A}$ is the set of (β, γ) *feasible control parameters*, defined by

$$\mathcal{A} = \{ (\beta, \gamma) \in \mathcal{P} \mid \mathbb{E} \int_0^T B(t, \beta_t, \gamma_t)^2 dt < +\infty \}$$

and where the ex-post price $(X^{\beta, \gamma}(v, \tilde{S}_v), \pi^{\beta, \gamma}(v, \tilde{S}_v))$ is, for each control process $(\beta, \gamma) \in \mathcal{A}$, the unique solution of the LBSDE associated with terminal time v , terminal condition $\tilde{S}_v$ and a linear generator; more precisely,

$$\begin{cases} -dX_t^{\beta, \gamma} &= (B(t, \beta_t, \gamma_t) - \beta_t X_t^{\beta, \gamma} - \gamma_t \pi_t^{\beta, \gamma}) dt - (\pi_t^{\beta, \gamma})^* dW_t \\ X_v^{\beta, \gamma} &= \tilde{S}_v. \end{cases} \quad (5.3)$$

Consequently, X_t is the value of the optimization problem

$$X_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} \operatorname{ess\,inf}_{(\beta, \gamma) \in \mathcal{A}} X_t^{\beta, \gamma}(v, \tilde{S}_v).$$

Then, applying the previous results on RBSDEs, it follows that the price $(X_t, 0 \leq t \leq T)$ corresponds to the solution of the RBSDE associated with terminal condition ξ , generator b and obstacle S , that is

Proposition 5.1 *There exists $(\pi_t) \in \mathbb{H}^2$ and (K_t) an increasing adapted continuous process with $K_0 = 0$ such that*

$$-dX_t = b(t, X_t, \pi_t)dt + dK_t - \pi_t^* dW_t \quad ; \quad X_T = \xi, \quad (5.4)$$

with $X_t \geq S_t$, $0 \leq t \leq T$, and $\int_0^T (X_t - S_t) dK_t = 0$.

Furthermore, the stopping time $D_t = \inf\{t \leq s \leq T; X_s = S_s\}$ is optimal that is

$$X_t = \operatorname{ess\,sup}_{v \in \mathcal{T}_t} X_t(v, \tilde{S}_v) = X_t(D_t, \tilde{S}_{D_t}).$$

The process K_t may be interpreted as a cumulative consumption process. Such a triple (Y_t, π_t, K_t) satisfying equation (5.4) with $Y_t \geq S_t$, $0 \leq t \leq T$ (but not necessarily $\int_0^T (Y_t - S_t) dK_t = 0$) is called a superhedging strategy for the American option (S, ξ) . Consequently, by the corollary of the comparison theorem 2.2 for RBSDEs, the price X_t is equal to the so-called ‘upper price’ defined as the smallest of the superhedging strategies for (S, ξ) .

References

- Barles, G. (1994) *Solutions de viscosité des équations de Hamilton–Jacobi du premier ordre et Applications*, Math. Appl **17**, Springer.
- Barles, G. & Burdeau, J. (1994) ‘The Dirichlet problem for semilinear second order degenerate elliptic equations and applications to stochastic exit time problems’, preprint.
- Bensoussan, A. (1984) ‘On the theory of option pricing’, *Acta Applicandae Mathematicae* **2**, 139–158.
- Bensoussan, A. & Lions, J.L. (1978) *Applications des Inéquations Variationnelles en Contrôle Stochastique*, Dunod.
- Crandall, M., Ishii, H. & Lions, P.L. (1992) ‘User’s guide to the viscosity solutions of second order partial differential equations’, *Bull. A.M.S.* **27**, 1–67.
- Darling, R. (1994) ‘Constructing gamma–martingales with prescribed limits using backward SDEs’, preprint.

- Davis, M. & Karatzas, I. (1994) 'A deterministic approach to optimal stopping,' in *Probability, Statistics and Optimization*, F.P. Kelly (ed.), 455–466, Wiley.
- Duffie, D. & Epstein, L. (1992) 'Stochastic differential utility', *Econometrica* **60**, 353–394.
- El Karoui, N. (1981) 'Les aspects probabilistes du contrôle stochastique', in *Ecole d'Été de Saint Flour*, LNM **876**, Springer.
- El Karoui, N. & Chaleyat-Maurel, M. (1978) 'Un problème de réflexion et ses applications au temps local et aux EDS sur $\mathbb{R}$, cas continu', in *Temps locaux*, Astérisque **2–53**, SMF.
- El Karoui, N. & Jeanblanc-Picqué, M. (1993) 'Optimization of consumption with labor income', preprint.
- El Karoui, N., Peng, S. & Quenez, M.C. (1994) 'Backward stochastic differential equations in finance', Preprint Université Paris VI, **260**.
- El Karoui, N. & Quenez, M.C. (1997) 'Imperfect markets and backward stochastic differential equations', this volume.
- El Karoui, N., Kapoudjian, C., Pardoux, E., Peng, S. & Quenez, M.C. (1995) 'Reflected solutions of backward SDEs, and related obstacle problems for PDEs', preprint.
- Hamadene, S. & Lepeltier, J.P. (1995) 'Zero-sum stochastic differential games and backward equations,' *Systems & Control Letters* **24**, 259–263.
- Jacka, S. (1993) 'Local times, optimal stopping and semimartingales', *Annals of Probability* **21**, 329–339.
- Karatzas, I. (1988) 'On the pricing of American options', *Appl. Math. Optimization* **17**, 37–6.
- Karatzas, I. (1989) 'Optimization problems in the theory of continuous trading', *SIAM. J. Control. Optim.* **27**, 1221–1259.
- Karatzas, I. & Shreve, S. (1996) *Methods of Mathematical Finance*, book in preparation.
- Pardoux, E. & Peng, S. (1990) 'Adapted solution of a backward stochastic differential equation', *Systems & Control Letters* **14**, 55–61.
- Pardoux, E. & Peng, S. (1992) 'Backward SDEs and quasilinear PDEs', in *Stochastic Partial Differential Equations and their Applications*, B.L. Rozovskii, R.B. Sowers (eds.), LNCIS **176**, Springer.
- Pardoux, E. & Peng, S. (1995) 'Some backward SDEs with non Lipschitz coefficients', Proc. Conf. Metz, to appear.
- Revuz, D. & Yor, M. (1994) *Continuous Martingales and Brownian Motion*, Springer.